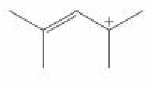
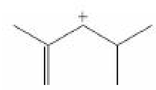
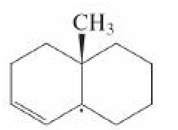
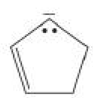
Problems

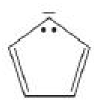
32. Draw all resonance forms and a representation of the appropriate resonance hybrid for each of the following species.

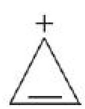
a. 

b. 

c. 

d. 

e. 

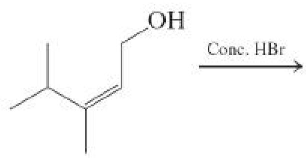
f. 

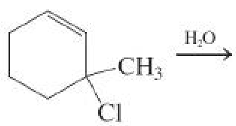
33. For each species in Problem 32, indicate the resonance form that is the major contributor to the resonance hybrid. Explain your choices.

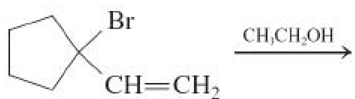
34. Illustrate by means of appropriate structures (including all relevant resonance forms) the initial species formed by **(a)** breaking the weakest C─H bond in 1-butene; **(b)** treating 4-methylcyclohexene with a powerful base (e.g., butyllithium-TMEDA); **(c)** heating a solution of 3-chloro-1-methylcyclopentene in aqueous ethanol.

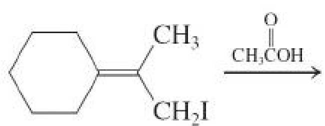
35. Rank primary, secondary, tertiary, and allylic radicals in order of decreasing stability. Do the same for the corresponding carbocations. Do the results indicate something about the relative ability of hyperconjugation and resonance to stabilize radical and cationic centers?

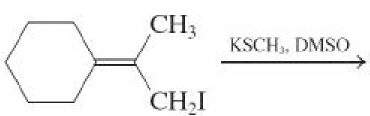
36. Give the major product(s) of each of the following reactions.

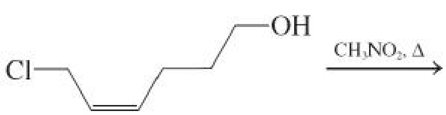
a. 

b. 

c. 

d. 

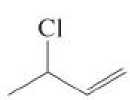
e. 

f. 

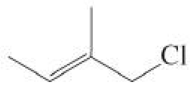
37. Formulate detailed mechanisms for the reactions in Problem 36, parts (a), (c), (e), and (f).

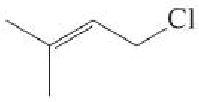
38. Rank primary, secondary, tertiary, and (primary) allylic chlorides in approximate order of **(a)** decreasing SN1S_N1 reactivity; **(b)** decreasing SN2S_N2 reactivity.

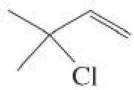
39. Rank the following six molecules in approximate order of decreasing SN1S_N1 reactivity and decreasing SN2S_N2 reactivity.

a. 

b. 

c. 

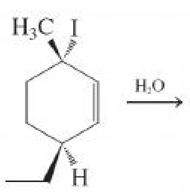
d. 

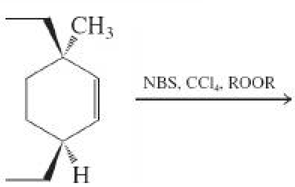
e. 

f. 

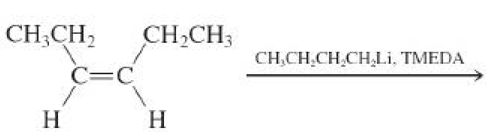
40. How would you expect the SN2S_N2 reactivities of simple saturated primary, secondary, and tertiary chloroalkanes to compare with the SN2 S_N2 reactivities of the compounds in Problem 39? Make the same comparison for SN1S_N1 reactivities.

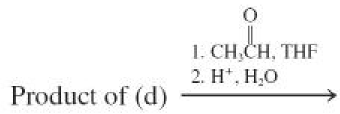
41. Give the major product(s) of each of the following reactions.

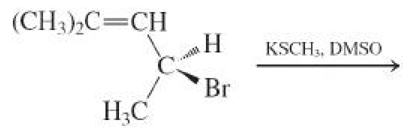
a. 

b. 

c. $(S)\text{-}CH_3CH_2\overset{\overset{CH_3}{|}}{C}HCH{=}CH_2 \xrightarrow{\text{NBS, } CCl_4\text{, ROOR}}$

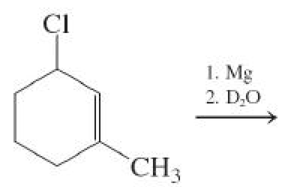
d. 

e. 

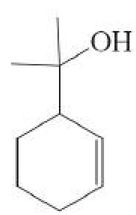
f. 

42. Write a detailed step-by-step mechanism to show how each of the products arises from the reaction in Problem 41, part (a).

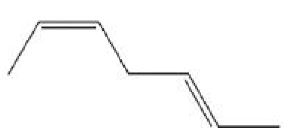
43. The following reaction sequence gives rise to two isomeric products. What are they? Explain the mechanism of their formation.



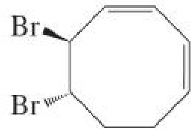
44. Starting with cyclohexene, propose a reasonable synthesis of the cyclohexene derivative shown below.

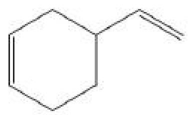


45. Give a systematic name to each of the following molecules.

a. 

b. 

c. 

d. 

46. Compare the allylic bromination reactions of 1,3-pentadiene and 1,4-pentadiene. Which should be faster? Which is more energetically favorable? How do the product mixtures compare?

CH2═CH—CH═CH—CH3 → NBS,ROOR,CCl4CH2═CH—CH2—CH═CH2 → NBS,ROOR,CCl4

$$CH_2\text{â}\,CH—CH\text{â}\,CH—CH_3 \xrightarrow{NBS,ROOR,CCl_4}$$

$$CH_2\text{â}\,CH—CH_2—CH\text{â}\,CH_2 \xrightarrow{NBS,ROOR,CCl_4}$$

47. We learned in Section 14-6 that electrophilic additions to conjugated dienes at low reaction temperatures give kinetic product ratios. Furthermore, these kinetic mixtures may change to mixtures with thermodynamic product ratios when the temperature is raised. Do you think that cooling the thermodynamic product mixture back to the original low reaction temperature will change it back to the original kinetic ratio? Why or why not?

48. Compare the addition of H+H^+ to 1,3-pentadiene and 1,4-pentadiene (see Problem 46). Draw the structures of the products. Draw a qualitative reaction profile showing both dienes and both proton addition products on the same graph. Which diene adds the proton faster? Which one gives the more stable product?

49. What products would you expect from the electrophilic addition of each of the following reagents to 1,3-cycloheptadiene? **(a)** HI; **(b)** Br2,Br_2, in H2O;H_2O; **(c)** IN3;IN_3; **(d)** H2SO4H_2SO_4 in CH3CH2OH.CH_3CH_2OH. (**Hint:** For b and c, see Exercise 14-13, specifically drawings A-C.)

50. Give the products of the reaction of *trans*-1,3-pentadiene with each of the reagents in Problem 49.

51. What are the products of reaction of 2-methyl-1,3-pentadiene with each of the reagents in Problem 49?

52. Write a detailed step-by-step mechanism for the formation of each of the products in Problem 51.

53. Give the products expected from reaction of deuterium iodide (DI) with **(a)** 1,3-cycloheptadiene; **(b)** *trans*-1,3-pentadiene; **(c)** 2-methyl-1,3-pentadiene. In what way does the observable result of reaction of DI differ from that of reaction of HI with these same substrates [compare with parts (a) of Problems 49 through 51]?

54. Arrange the following carbocations in order of decreasing stability. Draw all possible resonance forms for each of them.

a. CH2═CH−C+H2 $CH_2-CH-\overset{+}{C}H_2$

b. CH2═C+H CH_2â $\overset{+}{C}H$

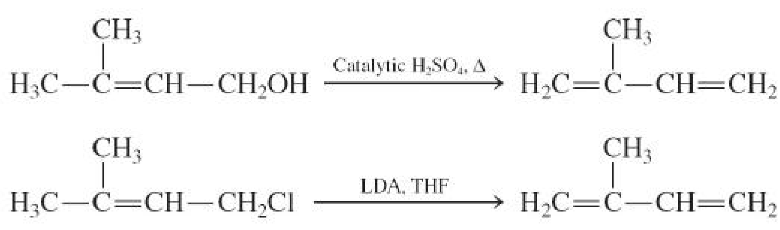
c. CH3C+H2 $CH_3\overset{+}{C}H_2$

d. CH3−CH═CH−C+H−CH3 CH_3-CHâ $CH-\overset{+}{C}H-CH_3$

e. CH2═CH−CH═CH−C+H2 CH_2â $CH-CH$â $CH-\overset{+}{C}H_2$

55. Sketch the molecular orbitals for the pentadienyl system in order of ascending energy (see Figures 14-2 and 14-7). Indicate how many electrons are present, and in which orbitals, for **(a)** the radical; **(b)** the cation; **(c)** the anion (see Figures 14-3 and 14-7). Draw all reasonable resonance forms for any one of these three species.

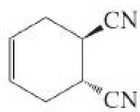
56. Dienes may be prepared by elimination reactions of substituted allylic compounds. For example,

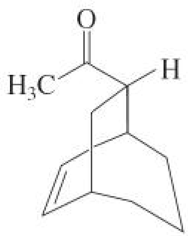


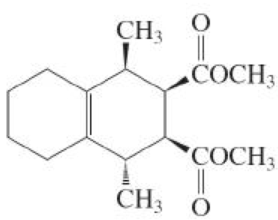
Propose detailed mechanisms for each of these 2-methyl-1,3-butadiene (isoprene) syntheses.

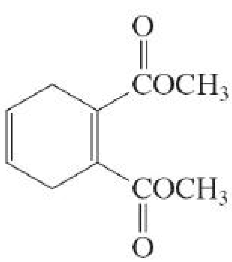
57. Give the structures of all possible products of the acid-catalyzed dehydration of vitamin A (Section 14-7).

58. Propose a synthesis of each of the following molecules by Diels-Alder reactions.

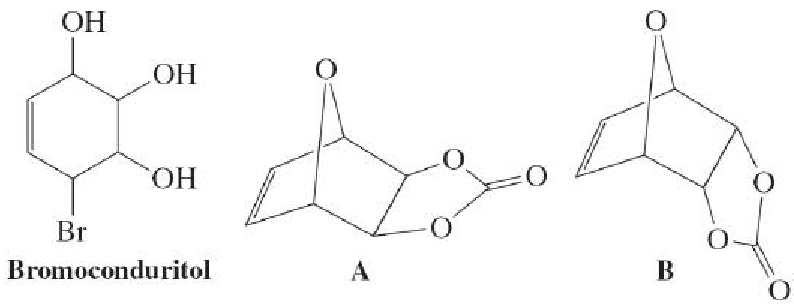
a. 

b. 

c. 

d. 

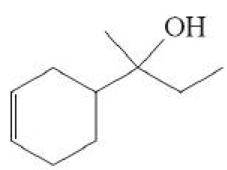
59. The haloconduritols are members of a class of compounds called glycosidase inhibitors. These substances possess an array of intriguing biological functions from antidiabetic and antifungal to activity against HIV virus and cancer metastasis. Stereoisomeric mixtures of bromoconduritol (below) are commonly used in studies of these properties. One recent synthesis of these compounds proceeds via the bicyclic ethers A and B. **(a)** Identify the starting materials for a one-step preparation of both of these ethers via a Diels-Alder reaction. **(b)** Which of the starting molecules in your answer is the diene, and which is the dienophile? **(c)** This Diels-Alder reaction gives an 80:20 mixture of B and A. Explain.



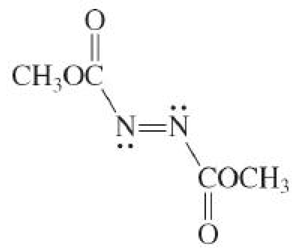
60. Give the product(s) of each of the following reactions.

- **a.** 3-Chloro-1-propene (allyl chloride) + NaOCH3$NaOCH_3$
- **b.** *cis*-2-Butene + NBS, peroxide (ROOR)
- **c.** 3-Bromocyclopentene + LDA
- **d.** *trans,trans*-2,4-Hexadiene + HClHCl
- **e.** *trans,trans*-2,4-Hexadiene + Br2,H2OBr_2, H_2O
- **f.** 1,3-Cyclohexadiene + methyl propenoate (methyl acrylate)
- **g.** 1,2-Dimethylidenecyclohexane + methyl propenoate (methyl acrylate)

61. **CHALLENGE** Propose an efficient synthesis of the cyclohexenol below, beginning exclusively with acyclic starting materials and employing sound retrosynthetic analysis strategy. [**Hint:** A Diels-Alder reaction may be useful, but take note of structural features in dienes and dienophiles that permit Diels-Alder reactions to work well (Section 14-8).]

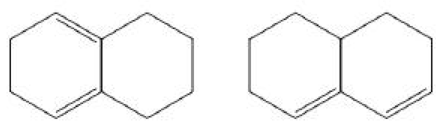


62. Dimethyl azodicarboxylate (see below) takes part in the Diels-Alder reaction as a dienophile. Write the structure of the product of cycloaddition of this molecule with each of the following dienes. **(a)** 1,3-Butadiene; **(b)** *trans, trans*-2,4-hexadiene; **(c)** 5,5-dimethoxycyclopentadiene; **(d)** 1,2-dime-thylidenecyclohexane. Ignore the stereochemistry at nitrogen in the products (amines experience rapid inversion, as we shall see in Section 21-2).



Dimethyl azodicarboxylate

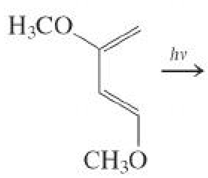
63. Bicyclic diene A reacts readily with appropriate alkenes by the Diels-Alder reaction, whereas diene B is totally unreactive. Explain.



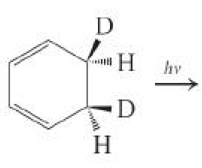
A **B**

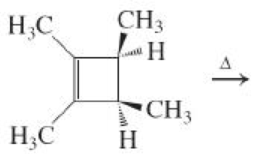
64. Formulate the expected product of each of the following conversions.

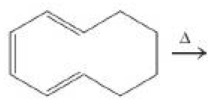
a.



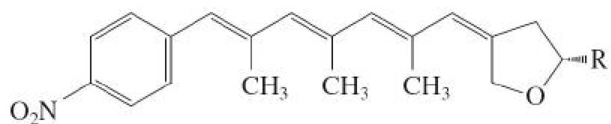
b.



c. 

d. 

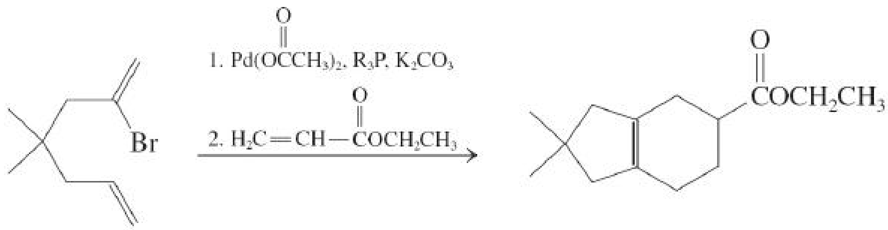
65. Microorganisms are a rich source of biologically active molecules with potentially useful medicinal activity. The *Streptomyces* family of bacteria is particularly active in this regard, producing a wide array of compounds including polyenes such as spectinabilin, an antiviral compound that contains the conjugated tetraene system shown below.



Partial structure of spectinabilin

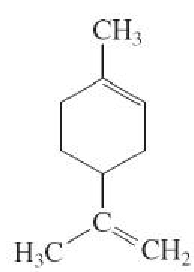
(a) Assign *E*,*Z* stereochemistry to the double bonds in the tetraene portion of spectinabilin. **(b)** Spectinabilin is yellow in color with $\lambda max=367$ nm$\lambda_{max} = 367\ \text{nm}$. Explain how this property is consistent with its partial structure. **(c)** One of the C−O bonds in the oxacyclopentane ring of spectinabilin is particularly prone to cleavage, opening the ring. Which one? Why? **(d)** Upon exposure to sunlight, the two central double bonds in spectinabilin exhibit photochemical geometric isomerism. Write the structure of the resulting product. **(e)** The product from part (d) spontaneously undergoes two successive thermal electrocyclic ring-closing reactions, the first involving eight ππ electrons and the second involving six. Using two-electron arrows, show the mechanisms of these processes and their products, the second of which is also powerfully biologically active. Are these reactions conrotatory or disrotatory?

66. Explain the following reaction sequence. (**Hint:** Remember the Heck reaction, Section 13-9.)



67. Give abbreviated structures of each of the following compounds: **(a)** (*E*)-1,4-poly-2-methyl-1,3-butadiene [(*E*)-1,4-polyisoprene]; **(b)** 1,2-poly-2-methyl-1,3-butadiene (1,2-polyisoprene); **(c)** 3,4-poly-2-methyl-1,3-butadiene (3,4-polyisoprene); **(d)** copolymer of 1,3-butadiene and ethenylbenzene (styrene, C6H5CH═CH2, C_6H_5CHâCH_2, SBR, used in automobile tires); **(e)** copolymer of 1,3-butadiene and propenenitrile (acrylonitrile, CH2═CHCN, CH_2â$CHCN$, latex); **(f)** copolymer of 2-methyl-1,3-butadiene (isoprene) and 2-methylpropene (butyl rubber, for inner tubes).

68. The structure of the terpene limonene is shown below (see also Solved Exercise 5-29). Identify the two 2-methyl-1,3-butadiene (isoprene) units in limonene. **(a)** Treatment of isoprene with catalytic amounts of acid leads to a variety of oligomeric products, one of which is limonene. Devise a detailed mechanism for the acid-catalyzed conversion of two molecules of isoprene into limonene. Take care to use sensible intermediates in each step. **(b)** Two molecules of isoprene may also be converted into limonene by a completely different mechanism, which takes place in the strict absence of catalysts of any kind. Describe this mechanism. What is the name of the reaction?



Limonene

69. **CHALLENGE** The carbocation derived from geranyl pyrophosphate (Section 14-10) is the biosynthetic precursor of not only camphor, but also limonene ([Problem 68]) and a-pinene ([Problem 49] of [Chapter 4]). Formulate mechanisms for the formation of the latter two compounds.

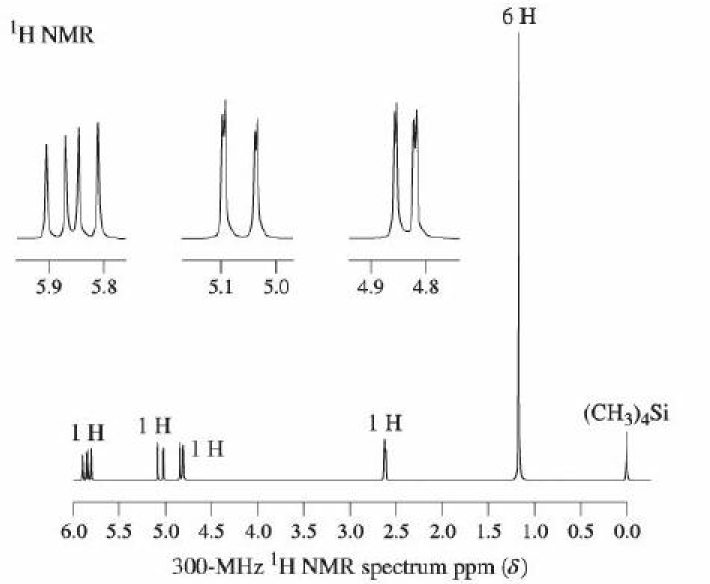
70. What is the longest-wavelength electronic transition in each of the following species? Use molecular-orbital designations, such as n → π* $n\rightarrow\pi^*$, π1 → π2$\pi_1\rightarrow\pi_2$, in your answer. (**Hint:** Construct a molecular-orbital energy diagram, like that in [Figure 14-16], for each.) **(a)** 2-Propenyl (allyl) cation; **(b)** 2-propenyl (allyl) radical; **(c)** formaldehyde, H2C═O;$H_2C\hat{a}\,O$; **(d)** N2;N_2; **(e)** pentadienyl anion ([Problem 55]); **(f)** 1,3,5-hexatriene.

71. Ethanol, methanol, and cyclohexane are commonly used solvents for UV spectroscopy because they do not absorb radiation of wavelength longer than 200 nm. Why not?

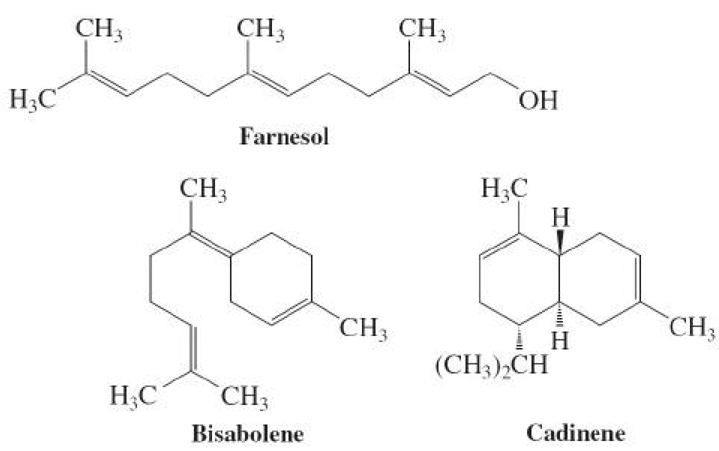
72. The ultraviolet spectrum of a 2×10−4$2\times10^{-4}$ M solution of 3-penten-2-one exhibits a π → π*$\pi\rightarrow\pi^*$ absorption at 224 nm with A=1.95$A=1.95$ and an n → π*$n\rightarrow\pi^*$ band at 314 nm with A=0.008$A=0.008$. Calculate the molar absorptivities (extinction coefficients) for these bands.

73. In a published synthetic procedure, acetone is treated with ethenyl (vinyl) magnesium bromide, and the reaction mixture is then neutralized with strong aqueous acid. The product exhibits the 1H NMR^{1}H NMR spectrum shown below and four signals in the 13C NMR^{13}C NMR (DEPT) spectrum, with δ=29.4(CH3),$\delta=29.4(CH_3)$, 71.1(Cquat), $71.1(C_{quat})$, 110.8(CH2),$110.8(CH_2)$, andand 146.3(CH) ppm. $146.3(CH)$ ppm. What is its structure? When the reaction mixture is (improperly) allowed to remain in contact with aqueous acid for too long, an additional new compound is observed. Its 1H NMR^{1}H NMR spectrum has peaks at δ=1.70 (s, 3 H),$\delta=1.70$ (s, 3 H), 1.79 (s, 3 H), 1.79 (s, 3 H), 2.25 (broad s, 1 H),2.25 (broad s, 1 H), 4.10 (d, J=8 Hz, 2 H),4.10 (d, $J=8$ Hz, 2 H), andand 5.45 (t, J=8 Hz, 1 H) ppm.5.45 (t, $J=8$ Hz, 1 H) ppm. Its 13C NMR ^{13}C NMR (DEPT) spectrum exhibits signals at δ=17.9(CH3),26.0(CH3),58.6(CH2), $\delta=17.9(CH_3), 26.0(CH_3), 58.6(CH_2)$, 121.0(CH),$121.0(CH)$, andand

134.7(Cquat) ppm. $134.7(C_{quat})$ ppm. What is the structure of the second product, and how did it get there?

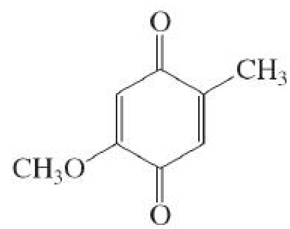


74. CHALLENGE Farnesol is a molecule that makes flowers smell good (lilacs, for instance). Treatment with hot concentrated H2SO4 H_2SO_4 converts farnesol first into bisabolene and finally into cadinene, an ingredient of the volatile oils of junipers and cedars. Propose detailed mechanisms for these conversions.



75. The ratio of 1,2- to 1,4-addition of Br2Br_2 to 1,3-butadiene (Section 14-6) is temperature dependent. Identify the kinetic and thermodynamic products, and explain your choices.

76. Diels-Alder cycloaddition of 1,3-butadiene with the cyclic dienophile shown below takes place at only one of the two carbon-carbon double bonds in the latter to give a single product. Give its structure and explain your answer. Watch stereochemistry.



This transformation was the initial step in the total synthesis of cholesterol (Section 4-7), completed by R. B. Woodward (see Section 14-9) in 1951. This achievement, monumental for its time, revolutionized synthetic organic chemistry.

Team Problems

77. As a team, consider the following historic preparation of a tris(1,1-dimethylethyl) derivative of Dewar benzene, B, by the photochemical isomerisation of 1,2,4-tris(1,1-dimethyle-thyl) benzene by van Tamelen and Pappas (1962). B does not revert to A via either a thermal or a photochemical electrocyclic mechanism. Formulate a mechanism for the conversion of A to B and explain the kinetic robustness of B with respect to the regeneration of A.